Lab 1

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1 Crafting A Compiler

1.1 Exercise 1.11 - MOSS

Answer:
MOSS convert source code into a token sequence. Then uses a "winnowing" algorithm to create fingerprints of the code by using a sliding window, hashing the code, and picking the minimum hash value. Matching fingerprints = similar code.

MOSS understands syntax of multiple programming languages. Which means it is able to recognize when code has been altered and restructured, but still remains functionally similar.

MOSS differs from alternatives like TurnItIn because others are mainly used for string matching. This fails with code because it is not plaintext, MOSS is able to understand structure and compare how similar they function. Changing a variable name will fool TurnItIn and competitors, but MOSS can see through that.

1.2 Exercise 3.1 - Token Sequence

Answer:
Source code:

```
main(){
 const float payment = 384.00;
 float bal;
 int month = 0;
 bal = 15000;
 while (bal>0){
  printf("Month: %2d Balance: %10.2f\n", month, bal);
  bal=bal-payment+0.015*bal;
  month=month+1;
 }
}
```

Token Sequence:

```
ID("main"), LPAREN, RPAREN, LBRACE,
 CONST, FLOAT, ID("payment"), ASSIGN, FLOAT_LITERAL(384.00), SEMICOLON,
 FLOAT, ID("BAL"), SEMICOLON
 INT, ID("month"), ASSIGN, INT_LITERAL(0), SEMICOLON,
 ID("BAL"), ASSIGN, INT_LITERAL(15000), SEMICOLON,
 WHILE, LPAREN, ID("BAL"), GT, INT_LITERAL(0), RPAREN, LBRACE
  ID("printf"), LPAREN, STRING_LITERAL("Month: %2d Balance: %10.2f\n"), COMMA, ID("month
  ID("BAL"), ASSIGN, ID("bal"), MINUS, ID("payment"), PLUS, FLOAT_LITERAL(0.015), MULT, I
  ID("month"), ASSIGN, ID("month"), PLUS, INT_LITERAL(1), SEMICOLON,
 RBRACE,
RBRACE
```

2 Dragon

2.1 Exercise 1.1.4

Answer:

Using C as a target language for source-to-source compilers holds several advantages: portability- C compilers are virtually everywhere optimization- So much work has been done on C, aka loads of optimization you dont need to do yourself Friendly- Use existing debugging tooling for c c is human-readable

2.2 Exercise 1.6.1

Answer:

- w = 13
- x = 11
- y = 13
- z = 11